The Chladni Plate

A brief introduction to some continuum mechanics and the chladni plate experiment

Vivek Kalita

Department of Physics, IISER Mohali

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1 Some Basics

1.1 Strain

Consider a continuous body occupying a region of three-dimensional space. Every material point in the body is specified by its position vector

$$\mathbf{r} = \begin{pmatrix} x \\ y \\ z \end{pmatrix}.$$

A deformation of the body can be described mathematically as a mapping in which every material point is assigned a new position. We write this mapping as

$$\mathbf{r} \longmapsto \mathbf{r} + \mathbf{u}(\mathbf{r})$$

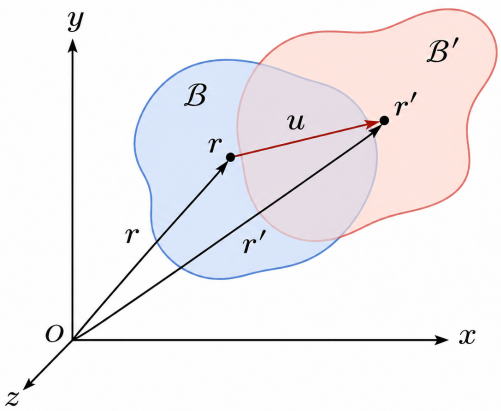


Figure 1: Deformation of a continous body

where $\mathbf{u}(\mathbf{r})$ is the displacement vector field, given by

$$\mathbf{u}(\mathbf{r}) = \begin{pmatrix} u_x(x,y,z) \\ u_y(x,y,z) \\ u_z(x,y,z) \end{pmatrix}.$$

1. Now consider an infinitesimal rectangular element of a continuous medium with dimensions dx, dy and dz.

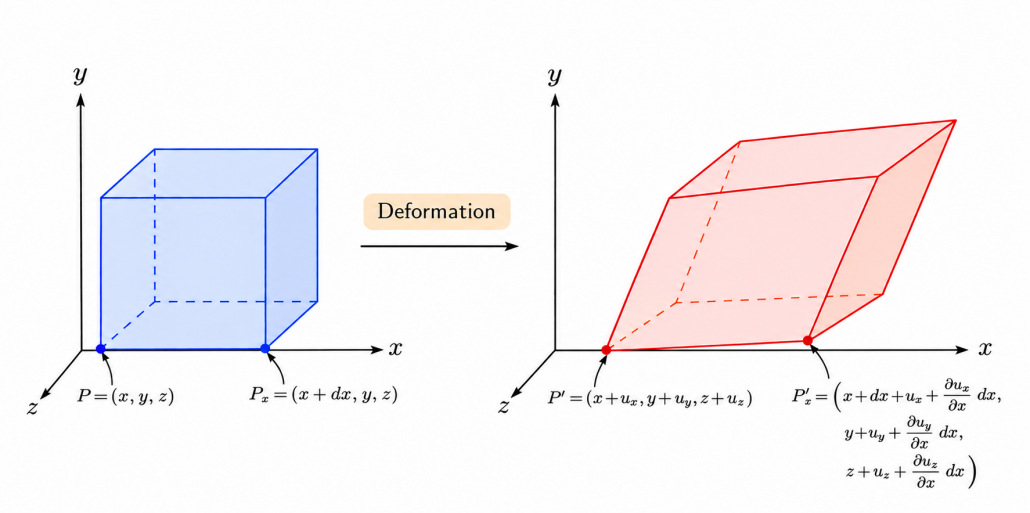


Figure 2: Deformations of a very small cuboid

Let one of its vertices be located at $P = (x, y, z)$. The three neighbouring vertices along the coordinate directions are initially located at

$$P_x = (x + dx, y, z), \qquad P_y = (x, y + dy, z), \qquad P_z = (x, y, z + dz).$$

Under a deformation, every material point is displaced according to the displacement field

$$\mathbf{u}(x,y,z) = (u_x(x,y,z),\ u_y(x,y,z),\ u_z(x,y,z))\,.$$

Consequently, the point P is mapped to

$$P' = (x + u_x(x,y,z),\ y + u_y(x,y,z),\ z + u_z(x,y,z))\,.$$

Similarly, the neighbouring point P_x is mapped to

$$P'_x = (x + dx + u_x(x + dx, y, z),\ y + u_y(x + dx, y, z),\ z + u_z(x + dx, y, z))\,.$$

Expanding the displacement components in a Taylor series about (x, y, z), and retaining only the first-order terms, we obtain the change in the position of the neighbouring point relative to P' to be:

$$P'_x - P' = \left(\left(1 + \frac{\partial u_x}{\partial x}\right) dx,\ \frac{\partial u_y}{\partial x} dx,\ \frac{\partial u_z}{\partial x} dx\right).$$

In particular, the change in the length of the edge along the x-direction, to first order, is

$$\Delta x = \frac{\partial u_x}{\partial x}\, dx.$$

2. Strain is defined as the fractional change in length, the normal strain along the x-direction is

$$\epsilon_{xx} = \frac{\Delta x}{dx} = \frac{\partial u_x}{\partial x}.$$

Following the same procedure for the edges along the y and z directions, we obtain

$$\epsilon_{yy} = \frac{\partial u_y}{\partial y}, \qquad \epsilon_{zz} = \frac{\partial u_z}{\partial z}.$$

3. From the previous calculations, we know that the change in height of the edge parallel to the x-axis is

$$\Delta y = \frac{\partial u_y}{\partial x}\, dx.$$

Therefore, for small deformations,

$$\alpha \approx \tan\alpha = \frac{\Delta y}{dx} = \frac{\partial u_y}{\partial x}.$$

Similarly, we can obtain

$$\beta \approx \tan\beta = \frac{\partial u_x}{\partial y}.$$

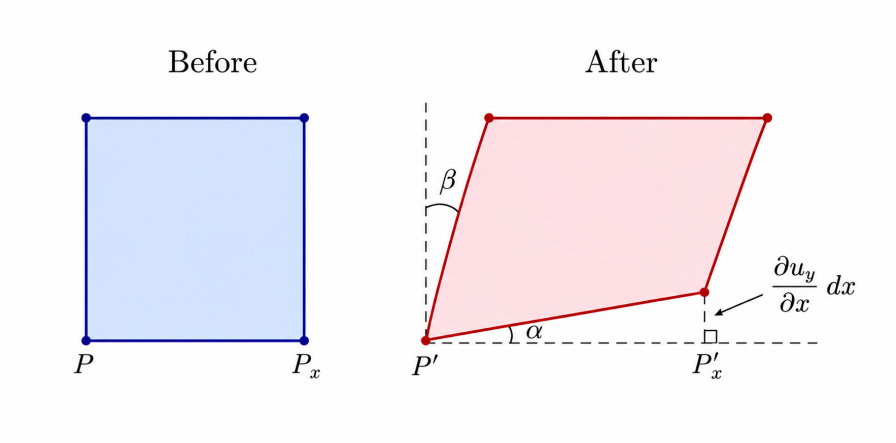


Figure 3: The shear strain

The total change in the right angle between the two edges is given by the sum of these two angles. Hence,

$$\gamma_{xy} = \alpha + \beta = \frac{\partial u_y}{\partial x} + \frac{\partial u_x}{\partial y}.$$

This quantity is called the engineering shear strain.

4. Therefore, the engineering shear strains are

$$\gamma_{xy} = \frac{\partial u_y}{\partial x} + \frac{\partial u_x}{\partial y} \quad \gamma_{yx} = \frac{\partial u_x}{\partial y} + \frac{\partial u_y}{\partial x}.$$

By convention, the components of the infinitesimal strain tensor are defined as half the engineering shear strain. Thus,

$$\epsilon_{xy} = \epsilon_{yx} = \frac{1}{2}\left(\frac{\partial u_y}{\partial x} + \frac{\partial u_x}{\partial y}\right).$$

Definition 1.1 (Strain Tensor).

The strain tensor is a second-order symmetric tensor that describes the deformation of a continuous body. It is given by

$$\boldsymbol{\epsilon} = \begin{pmatrix} \epsilon_{xx} & \epsilon_{xy} & \epsilon_{xz} \\ \epsilon_{yx} & \epsilon_{yy} & \epsilon_{yz} \\ \epsilon_{zx} & \epsilon_{zy} & \epsilon_{zz} \end{pmatrix}.$$

The diagonal components represent normal strains, given by

$$\epsilon_{xx} = \frac{\partial u_x}{\partial x} \quad \epsilon_{yy} = \frac{\partial u_y}{\partial y} \quad \epsilon_{zz} = \frac{\partial u_z}{\partial z}.$$

The off-diagonal components represent shear strains:

$$\epsilon_{xy} = \epsilon_{yx} = \frac{1}{2}\left(\frac{\partial u_x}{\partial y} + \frac{\partial u_y}{\partial x}\right), \quad \epsilon_{yz} = \epsilon_{zy} = \frac{1}{2}\left(\frac{\partial u_y}{\partial z} + \frac{\partial u_z}{\partial y}\right), \quad \epsilon_{xz} = \epsilon_{zx} = \frac{1}{2}\left(\frac{\partial u_x}{\partial z} + \frac{\partial u_z}{\partial x}\right).$$